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QUESTION

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QUESTION 1.2

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(users) XXXXXXXXXX XXXXX 2.1

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XXXXXX	XXXXXX	XXXXXXXX XXXX
id	INTEGER PK	XXXXXXXX XXXXXXX
username	VARCHAR(50) UNIQUE	XXXXXXXXXXXX XXXX
password_hash	VARCHAR(255)	XXXXXXXXXXXX XXXXXXX XXXXX
email	VARCHAR(100)	XXXXXXXXXXXX XXXXXXX
phone	VARCHAR(20)	XXXXXXXX XXXX
role	ENUM	XXXXXX
is_active	BOOLEAN	XXXXXXXX XXXX
created_at	TIMESTAMP	XXXXXXXX XXXXXXX
last_login	TIMESTAMP	XXXXX XXXXXXX XXXX

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(students) XXXXXXX XXXXX 2.2

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属性名	数据类型	完整性约束
id	INTEGER PK	NOT NULL, UNIQUE
user_id	INTEGER FK→users	NOT NULL
student_number	VARCHAR(20) UNIQUE	NOT NULL
full_name	VARCHAR(100)	NOT NULL
national_id	VARCHAR(20) UNIQUE	NOT NULL
date_of_birth	DATE	NOT NULL
gender	ENUM	NOT NULL
level	INTEGER	NOT NULL, CHECK (level IN (1, 2, 3, 4, 5))
major	VARCHAR(50)	NOT NULL
department_id	INTEGER FK→departments	NOT NULL
enrollment_status	ENUM	NOT NULL
gpa	DECIMAL(3,2)	NOT NULL, CHECK (gpa > 0)
cumulative_hours	INTEGER	NOT NULL, CHECK (cumulative_hours > 0)
advisor_id	INTEGER FK→professors	NOT NULL
enrollment_year	INTEGER	NOT NULL, CHECK (enrollment_year > 2000)
graduation_date	DATE	NOT NULL

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(departments) 2.3

[illegible]

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id	INTEGER PK	?????? ??????
name	VARCHAR(100)	????? ???
code	VARCHAR(10) UNIQUE	????? ???

hod_id	INTEGER FK→professors	INTEGER PK
contact_email	VARCHAR(100)	VARCHAR(100)
created_at	TIMESTAMP	TIMESTAMP

(professors) Database Design 2.4

. The database design for the professors table is as follows. The table has three columns: hod_id, contact_email, and created_at. The hod_id column is an integer foreign key that references the id column in the professors table. The contact_email column is a varchar(100) column. The created_at column is a timestamp column.

Column	DataType	Constraints
id	INTEGER PK	INTEGER PK
user_id	INTEGER FK→users	INTEGER FK→users
member_id	INTEGER FK→members	INTEGER FK→members
department_id	INTEGER FK→departments	INTEGER FK→departments
full_name	VARCHAR(100)	VARCHAR(100)
rank	ENUM	ENUM
specialization	VARCHAR(100)	VARCHAR(100)
qualification	VARCHAR(100)	VARCHAR(100)
max_hours_per_week	INTEGER	INTEGER
hire_date	DATE	DATE
is_active	BOOLEAN	BOOLEAN

(employees) Database Design 2.5

. The database design for the employees table is as follows. The table has three columns: id, user_id, and member_id. The id column is an integer primary key. The user_id column is an integer foreign key that references the id column in the users table. The member_id column is an integer foreign key that references the id column in the members table.

Column	DataType	Constraints
id	INTEGER PK	INTEGER PK
user_id	INTEGER FK→users	INTEGER FK→users

department_id	INTEGER FK→departments	INTEGER PK
full_name	VARCHAR(100)	VARCHAR(100)
position	VARCHAR(50)	VARCHAR(50)
permissions	JSON	JSON

(courses) Database Design for a University Database 2.6

The database design for the courses table is as follows. The table stores information about the courses offered by the university. The table has the following columns: id, code, name, credit_hours, semester_level, course_type, department_id, and description. The id column is the primary key. The code column is unique. The department_id column is a foreign key to the departments table. The description column is a text field.

Column	Data Type	Constraints
id	INTEGER PK	PRIMARY KEY
code	VARCHAR(20) UNIQUE	UNIQUE
name	VARCHAR(100)	
credit_hours	INTEGER	
semester_level	INTEGER	
course_type	ENUM	
department_id	INTEGER FK→departments	FOREIGN KEY
description	TEXT	

(prerequisites) Database Design for a University Database 2.7

The database design for the prerequisites table is as follows. The table stores information about the prerequisites for each course. The table has the following columns: id, course_code, prerequisite_code, and min_grade. The id column is the primary key. The course_code and prerequisite_code columns are foreign keys to the courses table. The min_grade column is a text field.

Column	Data Type	Constraints
id	INTEGER PK	PRIMARY KEY
course_code	VARCHAR(20) FK→courses	FOREIGN KEY
prerequisite_code	VARCHAR(20) FK→courses	FOREIGN KEY
min_grade	VARCHAR(5)	

(study_plans) Table Design 2.8

The study_plans table is used to store the information about the study plans. The table has the following fields:

- id: The primary key of the table, representing the unique identifier for each study plan.
- program_name: The name of the program, with a maximum length of 100 characters.
- department_id: The foreign key to the departments table, representing the department to which the study plan belongs.
- total_credit_hours: The total number of credit hours required for the study plan.
- level: The level of the study plan, such as undergraduate or graduate.
- academic_year: The academic year for which the study plan is valid.
- status: The status of the study plan, such as active or inactive.

Field Name	Field Type	Field Description
id	INTEGER PK	Primary key, unique identifier
program_name	VARCHAR(100)	Program name
department_id	INTEGER FK→departments	Department ID, foreign key to departments
total_credit_hours	INTEGER	Total credit hours required
level	INTEGER	Study level
academic_year	INTEGER	Academic year
status	ENUM	Status of the study plan

(plan_courses) Table Design 2.9

The plan_courses table is used to store the information about the courses in a study plan. The table has the following fields:

- id: The primary key of the table, representing the unique identifier for each course in a study plan.
- plan_id: The foreign key to the study_plans table, representing the study plan to which the course belongs.
- course_code: The foreign key to the courses table, representing the course to be taken.
- semester_order: The order of the course within the semester.
- course_type: The type of the course, such as lecture or lab.

Field Name	Field Type	Field Description
id	INTEGER PK	Primary key, unique identifier
plan_id	INTEGER FK→study_plans	Study plan ID, foreign key to study_plans
course_code	VARCHAR(20) FK→courses	Course code, foreign key to courses
semester_order	INTEGER	Semester order
course_type	ENUM	Course type

(sections) Table Design 2.10

The sections table is used to store the information about the sections of a course. The table has the following fields:

- id: The primary key of the table, representing the unique identifier for each section.
- course_code: The foreign key to the courses table, representing the course to which the section belongs.
- section_id: The foreign key to the sections table, representing the section to which the student belongs.
- section_order: The order of the section within the course.
- section_type: The type of the section, such as lecture or lab.

Attribute	Domain	Cardinality
id	INTEGER PK	1:1
course_code	VARCHAR(20) FK→courses	1:1
section_number	INTEGER	1:1
professor_id	INTEGER FK→professors	1:1
semester	ENUM	1:1
academic_year	INTEGER	1:1
capacity	INTEGER	1:1
enrolled_count	INTEGER	1:1
status	ENUM	1:1

(section_schedule) Table Structure 2.11

The section_schedule table stores the schedule of the sections. It contains the following attributes: id, section_id, day_of_week, start_time, end_time, and room_id. The id attribute is the primary key. The section_id attribute is a foreign key to the sections table. The day_of_week attribute is an enumeration type. The start_time and end_time attributes are time values. The room_id attribute is a foreign key to the rooms table.

Attribute	Domain	Cardinality
id	INTEGER PK	1:1
section_id	INTEGER FK→sections	1:1
day_of_week	ENUM	1:1
start_time	TIME	1:1
end_time	TIME	1:1
room_id	INTEGER FK→rooms	1:1

(rooms) Table Structure 2.12

The rooms table stores the information about the rooms. It contains the following attributes: id, name, and code. The id attribute is the primary key. The name attribute is a string. The code attribute is a string.

Attribute	Domain	Cardinality
id	INTEGER PK	1:1
name	VARCHAR(50)	1:1
code	VARCHAR(20) UNIQUE	1:1

building	VARCHAR(50)	varchar(50)
floor	INTEGER	integer
capacity	INTEGER	integer
room_type	ENUM	enum
equipment	JSON	json

(registrations) Database Design 2.13

The registrations table stores information about the registration of students in a course. It includes the student's ID, the section ID, the semester, the academic year, the registration date, the status, and the employee who approved the registration.

Field	Data Type	Constraints
id	INTEGER PK	primary key
student_id	INTEGER FK→students	foreign key to students
section_id	INTEGER FK→sections	foreign key to sections
semester	ENUM	enum
academic_year	INTEGER	integer
registration_date	TIMESTAMP	timestamp
status	ENUM	enum
approved_by	INTEGER FK→employees	foreign key to employees

(student_transcripts) Database Design 2.14

The student_transcripts table stores information about the transcripts of students. It includes the student's ID, the course code, the semester, and the grade.

Field	Data Type	Constraints
id	INTEGER PK	primary key
student_id	INTEGER FK→students	foreign key to students
course_code	VARCHAR(20) FK→courses	foreign key to courses
semester	ENUM	enum
grade	VARCHAR(5)	varchar(5)

grade_points	DECIMAL(3,2)	0.00-4.00
credit_hours	INTEGER	0-18
status	ENUM	Completed, In Progress, On Hold

(academic_advisors) Database Design for the University of Phoenix 2.15

The academic_advisors table stores information about the advisors assigned to each student. The table includes the advisor's ID, the professor's ID, the student's ID, the department's ID, and the date the advisor was assigned. The table is a many-to-many relationship between the professors and students tables.

Field	Data Type	Constraints
id	INTEGER PK	Primary Key
professor_id	INTEGER FK→professors	Foreign Key to professors
student_id	INTEGER FK→students	Foreign Key to students
department_id	INTEGER FK→departments	Foreign Key to departments
assigned_date	DATE	

(graduation_requests) Database Design for the University of Phoenix 2.16

The graduation_requests table stores information about the graduation requests for each student. The table includes the student's ID, the department's ID, the request date, the GPA at the time of the request, the completed hours, the missing courses, the department's approval, the hod's approval, and the date the request was approved. The table is a one-to-many relationship between the students and graduation_requests tables.

Field	Data Type	Constraints
id	INTEGER PK	Primary Key
student_id	INTEGER FK→students	Foreign Key to students
department_id	INTEGER FK→departments	Foreign Key to departments
request_date	DATE	
gpa_at_request	DECIMAL(3,2)	
completed_hours	INTEGER	
missing_courses	JSON	
department_approval	BOOLEAN	
hod_approved_by	INTEGER FK→professors	Foreign Key to professors
hod_approved_at	TIMESTAMP	

registration_approval	BOOLEAN	XXXXXXXXXX XXXXXXXX
registration_approved_by	INTEGER FK→employees	XXXXXXXXXX XX XXXXX
final_status	ENUM	XXXXXXXXXX XXXXXXXX
graduation_date	DATE	XXXXXXXXXX XXXXXX

(notifications) XXXXXXXXXXXX XXXXX 2.17

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XXXXXX	XXXXXX	XXXXXXXXXX XXXX
id	INTEGER PK	XXXXXXXXXX XXXXXXXX
user_id	INTEGER FK→users	XXXXXXXXXX XXXXX
title	VARCHAR(200)	XXXXXXXXXX
message	TEXT	XXXXXXXXXX
type	ENUM	XXXXXX
is_read	BOOLEAN	XXXXXXXXXX XXXX
created_at	TIMESTAMP	XXXXXXXXXX XXXXXXXX

(employee_transfers) XXXXXXXXXXXX XXXXXXXX XXXXX 2.18

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XXXXXX	XXXXXX	XXXXXXXXXX XXXX
id	INTEGER PK	XXXXXXXXXX XXXXXXXX
employee_id	INTEGER FK→employees	XXXXXXXXXX XXXXX
current_role	VARCHAR(50)	XXXXXXXXXX XXXXXXXX
requested_role	VARCHAR(50)	XXXXXXXXXX XXXXXXXX
requested_rank	VARCHAR(50)	XXXXXXXXXX XXXXXXXX
specialization	VARCHAR(100)	XXXXXX
courses_to_teach	JSON	XXXXXXXXXX XXXXXXXX XXXXXXXXXX

reason	TEXT	??????
status	ENUM	???????
reviewed_by	INTEGER FK→professors	????????
review_notes	TEXT	?????????? ????????

(audit_log) ?????????? ??? ????? 2.19

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属性名	属性型	属性制約
id	INTEGER PK	UNIQUE NOT NULL
action	VARCHAR(50)	NOT NULL
entity_type	VARCHAR(50)	NOT NULL
entity_id	INTEGER	NOT NULL
performed_by	INTEGER FK→users	NOT NULL
details	JSON	NOT NULL
created_at	TIMESTAMP	NOT NULL

(semesters) **???????? ???? ???? 2.20**

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 ۳. ۱۳۹۸/۰۱/۰۱ تا ۱۳۹۸/۰۱/۰۱
 ۴. ۱۳۹۸/۰۱/۰۱ تا ۱۳۹۸/۰۱/۰۱
 ۵. ۱۳۹۸/۰۱/۰۱ تا ۱۳۹۸/۰۱/۰۱
 ۶. ۱۳۹۸/۰۱/۰۱ تا ۱۳۹۸/۰۱/۰۱
 ۷. ۱۳۹۸/۰۱/۰۱ تا ۱۳۹۸/۰۱/۰۱
 ۸. ۱۳۹۸/۰۱/۰۱ تا ۱۳۹۸/۰۱/۰۱
 ۹. ۱۳۹۸/۰۱/۰۱ تا ۱۳۹۸/۰۱/۰۱
 ۱۰. ۱۳۹۸/۰۱/۰۱ تا ۱۳۹۸/۰۱/۰۱

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id	INTEGER PK	?????? ??????
name	VARCHAR(50)	????? ???
academic_year	INTEGER	?????????? ?????
start_date	DATE	???????? ?????
end_date	DATE	???????? ?????
registration_start	DATE	???????? ?????
registration_end	DATE	???????? ?????

add_drop_end	DATE	XXXXXXXXXX XXXXXXXX XXXXXX
status	ENUM	XXXXXXXX

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QUESTION: What is the difference between a **strong** and a **weak** password?

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 0000000 000000 000000000 00 000 0000 0000 0000000000000 0000 .0000000000
 0000000000 0000 000000 00000000000 0000 0000000 00 00000000000
 000000000000000 0000000 000 0000 000000 000 000000

3.1

 Türkiye Cumhuriyeti Millî Eğitim Bakanlığı

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(section_id) 00000000 00000000 00000.(student_id) 00000000 00000:00000000
00000000 00000000 00 00000000 00000000 00000 000000000:1 00000000
(00000000 000000 00000) 00000000 00000000 0000000000 00000 000000000:2 00000000
:000000 000000 0000:3 00000000
00000 00000000 00000000 0000000000 00000 000000000:0.3
:00000000 00000000 00000 00 0000 0000:0.3
:000000000 00000000 00000 00 00000000 00000000 0000 0000 -
(00000000,10000000)0000000 > (00000000,10000000)0000000:00000000 0000000000 0000 -
00000000000 00 00000000 00000000 ← 00000000 0000 0000 -
0000000 000000 0 00000000:4 00000000
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3.2

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[illegible][illegible][illegible]

QUESTION 3.3

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    000000,000000,000000,000000,(room_id) 0000000 00000 :00000000
    0000000 0000000 0000000 000 00 000000 000000 000000000 :1 0000000
    :000000 00000 000 :2 0000000
    (2000000,1000000)000000 > (2000000,1000000)000000 :0000000 000000000 000 :0.2
    000000 000000 ← 000000 000 000 :0.2
    000000 00000 0 000000 :3 000000

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                                0000 000000 :3 000000

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[illegible]

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QUESTION: What is the difference between a **strong** and a **weak** password?

QUESTION

QUESTION 2: What is the probability that the first two cards are aces and the last two are kings?

QUESTION 10: What is the correct answer?

QUESTION 4: What is the difference between a **strong** and a **weak** password?

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QUESTION: What is the difference between a **strong** and a **weak** password?

QUESTION

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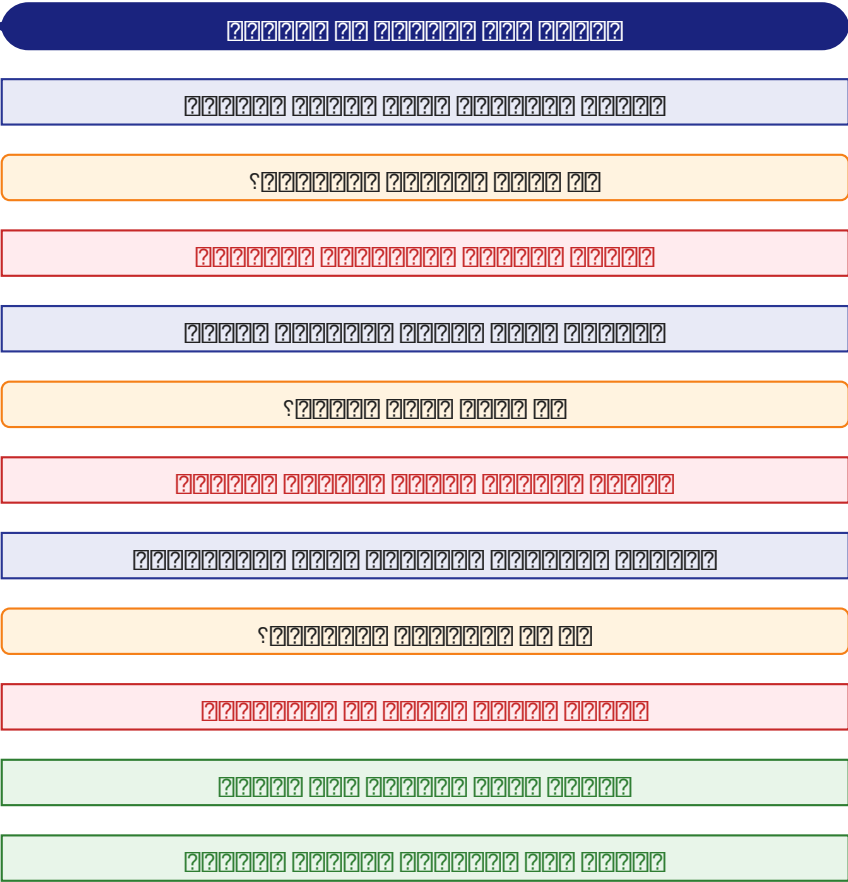
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QUESTION: What is the difference between a **strong** and a **weak** password?

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QUESTION	ANSWER	EXPLANATION
1. Which of the following is a characteristic of a primary cell?	4.00 2.00	Primary cells are designed for single use and cannot be recharged. They typically have a higher internal resistance compared to secondary cells.
2. A cell with an EMF of 1.5 V and an internal resistance of 0.5 Ω is connected to a load resistor. The terminal voltage across the load is 1.2 V. What is the resistance of the load?	(1.5 V / (1.5 V - 1.2 V)) * 0.5 Ω = 12.5 Ω	Using the voltage divider rule, the terminal voltage V_t = EMF * R_load / (R_load + r). Rearranging for R_load gives R_load = (EMF * r) / (EMF - V_t).
3. A cell with an EMF of 1.5 V and an internal resistance of 0.5 Ω is connected to a load resistor. The current flowing through the circuit is 0.5 A. What is the power dissipated in the load resistor?	Power = I^2 * R_load = (0.5 A)^2 * 12.5 Ω = 3.125 W	First, find the load resistance R_load using the terminal voltage and current. Then, use the formula P = I^2 * R to calculate the power dissipated in the load.
4. A cell with an EMF of 1.5 V and an internal resistance of 0.5 Ω is connected to a load resistor. The power dissipated in the load resistor is 3.125 W. What is the terminal voltage across the load?	Power = I^2 * R_load = 3.125 W. EMF = 1.5 V. Internal resistance r = 0.5 Ω. Find I and then V_t.	Use the power formula P = I^2 * R to find the current I. Then, use the terminal voltage formula V_t = EMF - I * r to find the terminal voltage.
5. A cell with an EMF of 1.5 V and an internal resistance of 0.5 Ω is connected to a load resistor. The terminal voltage across the load is 1.2 V. What is the efficiency of the cell?	Efficiency = (V_t / EMF) * 100% = (1.2 V / 1.5 V) * 100% = 80%	Efficiency is the ratio of the terminal voltage to the EMF, expressed as a percentage.
6. A cell with an EMF of 1.5 V and an internal resistance of 0.5 Ω is connected to a load resistor. The current flowing through the circuit is 0.5 A. What is the power dissipated in the internal resistance?	Power = I^2 * r = (0.5 A)^2 * 0.5 Ω = 0.125 W	Use the formula P = I^2 * R to calculate the power dissipated in the internal resistance r.
7. A cell with an EMF of 1.5 V and an internal resistance of 0.5 Ω is connected to a load resistor. The power dissipated in the internal resistance is 0.125 W. What is the terminal voltage across the load?	Power = I^2 * r = 0.125 W. EMF = 1.5 V. Internal resistance r = 0.5 Ω. Find I and then V_t.	Use the power formula P = I^2 * R to find the current I. Then, use the terminal voltage formula V_t = EMF - I * r to find the terminal voltage.
8. A cell with an EMF of 1.5 V and an internal resistance of 0.5 Ω is connected to a load resistor. The terminal voltage across the load is 1.2 V. What is the maximum power that can be delivered to the load?	Maximum power occurs when R_load = r = 0.5 Ω. P_max = (EMF^2) / (4 * r) = (1.5 V)^2 / (4 * 0.5 Ω) = 1.125 W	The maximum power transfer theorem states that maximum power is delivered to the load when the load resistance is equal to the internal resistance of the source.
9. A cell with an EMF of 1.5 V and an internal resistance of 0.5 Ω is connected to a load resistor. The maximum power that can be delivered to the load is 1.125 W. What is the terminal voltage across the load?	Maximum power occurs when R_load = r = 0.5 Ω. P_max = 1.125 W. Find V_t.	Use the maximum power transfer theorem to find the load resistance R_load. Then, use the terminal voltage formula V_t = EMF - I * r to find the terminal voltage.
10. A cell with an EMF of 1.5 V and an internal resistance of 0.5 Ω is connected to a load resistor. The terminal voltage across the load is 1.2 V. What is the ratio of the power dissipated in the load to the power dissipated in the internal resistance?	Ratio = (V_t / EMF)^2 = (1.2 V / 1.5 V)^2 = 0.64	The ratio of the power dissipated in the load to the power dissipated in the internal resistance is equal to the square of the ratio of the terminal voltage to the EMF.

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